Feasibility-First Constrained Quality-Diversity Evolution of a Sub-50 g Printable Titanium Karambit

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***Abstract*—We present G4 (v4.0.0), a clean-slate architectural upgrade of the Jarambit evolutionary knife-design engine that resolves the representation, physics, and optimization bottlenecks identified in the G3 v3.0.0 retrospective audit. Whereas G3 ranked designs by an unconstrained weighted sum over 179 genes (31 of them inactive) and produced a 67.97 g finished (71.09 g as-printed) part with two structural safety-factor shortfalls (solid tip** SF = 0.431 **under a global-chord 1.82 N·m pry and ring-clamped cantilever grip spine** SF = 0.515**), G4 introduces a declarative 130-gene schema (genome/G4.0.toml, zero dead genes) coupled to a true-3D curvilinear implicit kernel, a compliant four-finger Winkler hand-contact equilibrium model (HM-1), and a lexicographic feasibility-first quality-diversity search (W_ERGOLIGHT_v2) using a Feasible/Infeasible Two-Population (FI-2Pop) Lloyd-relaxed CVT archive with a four-emitter bandit (Sep-CMA-MAE, Lamarckian constraint-repair, feasible mass-descent, and module-aware Iso+LineDD). Every certification gate is fail-closed via G4ReleaseAdmissionContract: three-level Roache grid-convergence index** ($\mathrm{GCI}_{21} = 0.37\,\% \le 5\,\%$)**, per-load-case failure sequence ledger** ($\min \mathrm{seq} = 2.17 \ge 1.10$ **across lateral, torsion, and spine baton), frame-invariant TET10 solid-tip FE with a local tip-bisector diamond crown** ($\mathrm{SF}_{\mathrm{LCB}} = 3.29 \ge 1.25$)**, 2D quadratic T6 serration tooth checks across all teeth including end teeth** (4/4 **inner and** 5/5 **spine teeth,** $\mathrm{SF}_{\mathrm{LCB}} = 12.07 \ge 1.25$)**, corrected Goodman fatigue** ($\mathrm{SF}_{\mathrm{LCB}} = 2.31 \ge 1.10$)**, and resolution-invariant morphological erosion powder drainability. Across a multi-seed cohort and five-arm ablation study (E0–E5), full G4 (E3) achieves a median score of** 91.84 **(IQR** 0.13**) and certifies winner `g4v400_s042_62763aa2` at 35.42 g finished mass (35.68 g as printed with bevel grind allowance; 67.3 % hollowness vs. its 108.30 g solid envelope), scoring 92.45 under W_ERGOLIGHT_v2 (94.90 on legacy W_ERGOLIGHT_v1) with zero constraint violations** ($V(x) = 0$) **across all 29 gates.**

***Index Terms*—Constrained quality diversity, FI-2Pop, MAP-Elites, CMA-MAE, additive manufacturing, Ti-6Al-4V, finite elements, Roache GCI, Winkler foundation.**

1 Introduction and Retrospective Audit of G3

Designing a one-piece, hollow Ti-6Al-4V karambit for laser powder-bed fusion (L-PBF) requires simultaneous satisfaction of coupled geometric, ergonomic, structural, and additive-manufacturing constraints. In our preceding G3 v3.0.0 release, an eight-island MAP-Elites search evolved a double-edged, double-serrated claw blade with an internal triply periodic minimal surface (TPMS) lattice. A post-release red-team audit of the G3 codebase, physics pipeline, and release receipt exposed six fundamental engineering lessons that motivate G4:

1) **Soft scalarization permits structural shortfalls.** In W_ERGOLIGHT_v1, all 21 objective terms were combined linearly through a soft knee function sfk without a feasibility barrier. Because mass carried weight $w = 0.370$ while individual structural terms carried 0.0174–0.0700, the optimizer rationally sacrificed safety factors on the solid tip (SF = 0.431) and spine-loaded grip (SF = 0.515) whenever thinning the geometry gained more mass points than it lost in structural points.
2) **Boundary-condition mismatch on the grip and tip.** G3 evaluated the 3,427 N spine baton impulse on a hollow grip cantilevered from the pommel ring ($x \approx 9.9$ mm), producing an unphysical ∼55 mm moment arm that does not exist when the knife is held in a closed fist supported by the palm and four fingers [12], [13]. Similarly, G3 applied thefull 1.82 N·m body torque to the distal 3 mm of the point about the global origin-to-tip chord rather than the local tip bisector.
3) **Coupled point planform and spine thickness.** In G3, the distal width profile $w(s)$ blended into the body width via a polynomial ramp over 4 mm, coupling the 3 mm planform point angle ($\approx 92^\circ$ on the v3 winner) to the spine thickness knots and preventing an acute $\le 55^\circ$ point from coexisting with a torsion-resistant spine.
4) **Surrogate fidelity gaps.** In G3's search loop, the plate mesh mesh_domain(&b_outer, &[], 0.9) ignored through-blade fuller windows, the three-level Roache grid-convergence index (GCI) [6] was held at a constant placeholder 0.50, the Goodman fatigue formula divided by capacity instead of multiplying by unit stress, the serration check evaluated only $N - 1$ interior teeth between N scallops, and voxel flood-fill drainability at $h = 0.40$ mm passed single-voxel diagonal necks that close at finer print resolutions.
5) **Genome bloat and representation ceiling.** Of the 179 genes in G3.0, 26 genes (including legacy CPPN weights and fixed parameters) were never read by the decoder, 5 self-adaptive σ genes were hashed into genome_sha256, and the blade mid-surface was strictly planar ($z \equiv 0$).
6) **Uniform-skin mass floor.** Using a uniform ∼1.0 mm grip skin, an unperforated longitudinal web, and a solid pommel ring rim imposed a ∼60–68 g mass floor.

G4 (crates/g4) is engineered from the ground up to resolve every one of these findings while preserving the historical v1/v2 and v3.0.0 artifacts byte-for-byte intact.

2 Declarative G4.0 Genome and True-3D Implicit Kernel

2.1 Single-Source Declarative Schema (genome/G4.0.toml)

G4 replaces hand-maintained index arrays with a single declarative schema genome/G4.0.toml compiled into Rust via

Release v4.0.0, schema hash `f012a915`, decoder hash `009b7ec5`. Every measured number in this paper is generated from evidence/g4/v4.0.0/release.json and evolution_summary.json by g4-release-tex.

build.rs. The schema defines **130 live phenotype genes** (116 continuous F64 and 14 discrete/categorical I64 genes; **zero dead genes** and **zero search-strategy genes** in the phenotype hash) partitioned into nine anatomical modules (Table 1):

- **Module H (Hand & Grip, 24 genes):** Anthropometric hand scale, superelliptic grip cross-sections, palm swell, finger choil depths, and flange-graded skin thickness law $t_{\text{skin}}(\theta, u) = t_{\text{base}} + c_2 \cos(2\theta) + s_1 \sin\theta + a_x(u - 0.5)$ (**Lever L-1**).
- **Module G (Ring & Guard, 16 genes):** Ergonomic finger-ring lip crown, neck fillet radius, drained hollow ring gallery fraction ring_hollow_t (**Lever L-4**), and parametric B-spline guard contours.
- **Module P (Blade Planform, 16 genes):** Log-spiral curvature progression, chord reach target, and a decoupled linear-wedge point planform law over $[0, L_{\text{tip,plan}}]$ parameterized directly by the true planform point angle $\theta_{\text{tip}} \in [32°, 48°]$.
- **Module Z (3D Curvilinear Camber & Twist, 8 genes):** Out-of-plane blade mid-surface camber $z_c(s)$ and axial pre-twist $\psi(s)$ framed via parallel-transport (Bishop) frames along the 3D spine curve $\gamma(s) \in \mathbb{R}^3$.
- **Module X (Blade Section, Bevels & Crown, 18 genes):** Core thickness, bi-concave fuller grooves (**Lever L-5**), primary bevel angles (17.5°–22.5° per side), apex flats (0.05–0.11 mm), and the distal tip-bisector diamond crown reinforcement t_tip_crown_mm over $a \in [1.5, L_{\text{crown}}]$ mm (**Lever L-8**).
- **Module I (Internal Lattice, Webs & Bulkheads, 18 genes):** Spatially graded Gyroid/Schwarz-D Sheet-TPMS relative densities (**Lever L-3**, **L-6**), perforated Warren web hole fraction rib_hole_frac (**Lever L-2**), transverse finger-station bulkheads, and powder-removal port diameters (≥ 2.50 mm).
- **Module E (Dual-Edge Serrations, 12 genes):** Scallop counts, depths, gullet duty factors, and plunge offsets on both inner and spine cutting edges.
- **Module T (Alien Trabecular & Exoskeletal Architecture, 10 genes):** Continuous gate alien_gate, tri-arched Bishop-frame neck-to-blade trabecular struts, and neutral-axis flank windowing.
- **Module A (Zero-Mean Surface Texture, 8 genes):** Knurl/Voronoi/flute blend weights and amplitude (0.10–0.38 mm) enforced as a strictly zero-mean displacement ($\int h_{\text{tex}}\, dA = 0$, **Lever L-7**).

2.2 Decoupled Point Wedge and Tip-Bisector Diamond Crown

In G4, the planform half-width $w(a)/2$ over the distal tip zone $a \in [0, L_{\text{tip,plan}}]$ follows an exact linear wedge $w_{\text{wedge}}(a) = 2r_{\text{tip}} + 2a\tan(\theta_{\text{tip}}/2)$ before transitioning via a C^2 quintic smoothstep ($\tau^3(10 - 15\tau + 6\tau^2)$ over $a \geq 3.2$ mm) into the body width law. Consequently, the planform point angle at both $a = 1.0$ mm and $a = 3.0$ mm equals θ_{tip} ($41.38° \leq 55°$ on the winner) independently of spine thickness. Meanwhile, the spine thickness over $a \in [1.5, L_{\text{crown}}]$ receives a local diamond-ridge crown $\Delta t_{\text{crown}}(a) \propto \sqrt{\sin(\pi u_c)}\, t_{\text{tip,crown}}$, raising the polar moment of inertia $J \propto w\, t_{\max}^3$ by $> 2.4\times$ at the cost of only +0.19 g of distal titanium (evidence/g4/studies/e06_tip_crown.json).

Table 1 | Declarative G4.0 genome modules (130 live phenotype genes, zero dead genes)

Mod.	Anatomical Subsystem	Cont.	Disc.	Primary Mass / Structural Role
H	Hand & Flange-Graded Grip	24	0	Lever L-1, L-7 (I_{zz} +38 %, flank relief)
G	Drained Ring & Guard	16	0	Lever L-4 (hollow ring gallery, fillet K_t)
P	Log-Spiral Planform & Point	16	0	Gate S4 ($\theta_{\text{tip}} \leq 55°$ at 1 & 3 mm)
Z	3D Camber & Twist Frame	8	0	Bishop parallel-transport 3D mid-surface
X	Section, Fuller & Tip Crown	18	0	Lever L-5, L-8 (diamond tip crown)
I	TPMS, Warren Web & Ports	14	4	Lever L-2, L-3, L-6 (perforated web, drain
E	Dual-Edge Serration Law	10	2	All teeth checked (N/N including ends)
T	Trabecular Exo-Architecture	6	4	Tri-arch Guard-Neck bridge, flank windov
A	Zero-Mean Grip Texture	4	4	Lever L-7 ($\int h_{\text{tex}}\, dA = 0.0$ g)
Total (G4.0.toml)		**116**	**14**	**130 live genes (schema_sha256: `f012a`**

3 Requirements, Hand Contact Model, and Constrained Objective

3.1 Machine-Readable Specification (RS-karambit-v4.toml)

All 29 hard gates, material constants, and ergonomic target bands are loaded from RS-karambit-v4.toml with `#[serde(deny_unknown_fields)]` and traceable provenance in PROVENANCE.md. For each structural gate $j \in \{1, \ldots, 17\}$, G4 evaluates the uncertainty-bounded lower confidence bound (LCB) safety factor:

$$\text{SF}_{j,\text{LCB}}(x) = \frac{\text{SF}_j(x)}{1 + k_{\text{LCB}}\sqrt{\text{GCI}_j^2 + \delta_{\text{mat}}^2}}, \qquad k_{\text{LCB}} = 1.645, \quad (1)$$

where GCI_j is the measured three-level Roache grid-convergence index and $\delta_{\text{mat}} = 0.022$ is the HIP L-PBF Ti-6Al-4V ELI yield scatter [14].

3.2 Unified Winkler Hand Contact Equilibrium (HM-1)

Rather than clamping the pommel ring in space while applying a 3,427 N baton impulse to the blade, G4 models the closed-fist human grip as a four-station compliant Winkler foundation (eval_g4_hand_model_hm1) at the little/ring, middle, index-groove, and index-choil/palm-heel stations $x_i \in \{18, 39, 60, 81\}$ mm [12], [13]. Solving the 2×2 rigid-body Winkler equilibrium for each external wrench $(\mathbf{F}_{\text{ext}}, \mathbf{M}_{\text{ext}})$ satisfies force and moment balance to $< 10^{-12}$ N and distributes the load across the inter-finger saddle span ($l_{\text{HM1}} \approx 14.5$ mm), eliminating the artificial 55 mm ring-clamp cantilever lever arm and raising the spine-baton sequence ratio from 0.515 (v3 ring clamp) to ≥ 1.62 (evidence/g4/studies/e01_v3_rescore.json).

3.3 Lexicographic Feasibility-First Objective (W_ERGOLIGHT_v2)

G4 replaces unconstrained linear weighting with Deb's feasibility-first lexicographic ordering [4]:

$$W_{\text{ERGOLIGHT_v2}}(x) = \begin{cases} -1000\, V(x) + 50\,(1 - \tilde{V}(x)), \\ 50 + 50\,(0.62\, S_{\text{mass}}(x) + 0.23\, E(x) + 0.15\, S_{\text{margin}}(x) \end{cases} \quad (2)$$

where $V(x) = \sum_{j=1}^{29} \max(0, g_j(x))^2$ is the total squared normalized constraint violation, $S_{\text{mass}}(x) = \text{clip}((90 - m_{\text{finished}}(x))/(90 - 28), 0, 1)$ rewards low finished mass, $E(x) = \prod_{k=1}^{8} e_k(x)^{\beta_k}$ is the eight-factor anthropometric ergonomic geometric mean, and $S_{\text{margin}}(x)$ is a saturating reserve bonus

over all 17 structural LCB margins. Any feasible design ($V(x) = 0$, score $\in [50, 100]$) strictly dominates every infeasible design ($V(x) > 0$, score < 50).

4 Calibrated Multi-Fidelity Physics and Verification

Every surrogate and certification solver in crates/g4/src/physics.rs is verified by automated tests in vv_suite.rs and R&D study receipts in evidence/g4/studies/:

1) **Sub-voxel smoothed-Heaviside volume integration (T-MASS-1, Study E-02):** Evaluating $H_\epsilon(\phi)$ with 8-point Gauss-Legendre sub-cell quadrature where $|\phi| \le \sqrt{3}h/2$ achieves 0.002 % volume error on a $R = 10$ mm sphere and 0.044 % on a torus at $h = 0.40$ mm (< 0.25 % drift between $h = 0.40$ mm and $h = 0.20$ mm).
2) **Resolution-invariant erosion drainability (T-DRAIN -1, Study E-02):** Cavity air is eroded by the physical powder-clearance ball radius $r_{\text{erode}} = 0.50$ mm ($\phi_{\text{void}} \le -(r_{\text{erode}} + 0.5h)$) before 6-connected flood fill, guaranteeing identical drainability classification across $h \in \{0.40, 0.20, 0.10\}$ mm.
3) **Hole-aware plate FE and 3-level Roache GCI (T-PLATE-1..2):** Through-blade fenestrations are subtracted from the CST+DKT sandwich plate mesh [7], unreferenced interior hole vertices are locked, and lateral capacity is evaluated across three refinement levels ($A_{\max} \in \{2.00, 1.00, 0.50\}\,\text{mm}^2$, $r = \sqrt{2}$) to compute a real Roache $\text{GCI}_{21} = 0.37$ % (observed order $p = 2.00$) admitted by PlateGciAdmission [6].
4) **Frame-invariant TET10 solid-tip FE (T-TIP-1..3, Study E-06):** The distal 28 mm tip submodel is transformed into the local tip-bisector frame with cell-centred thickness sampling and evaluated under LC-TIP-1 (0.80 N·m local tip-bisector torsion) and LC-TIP-2 (60 N lateral tip pry), achieving 0.000 % rigid-rotation variance.
5) **All-tooth T6 serration evaluation and corrected Goodman fatigue (T-SERR-4):** Every serration tooth $i \in \{0, \dots, N-1\}$ on both edges—including both end teeth—is evaluated under combined 180 N snag and 60 N lateral pry with Peterson notch sensitivity K_f [11] and local thickness knockdown $k_t(t_{\text{local}})$, while Goodman fatigue at $R = 0$ is computed from true root bending stress $\sigma_{\max} = 180\,\sigma_{1\text{N}}$.
6) **Locking-free B-bar hex8 voxel FE (T-VFE-1..4):** Selective-reduced / Wilson-condensed B-bar brick elements [8] pass the constant-strain patch test to 1.1×10^{-16} and reduce thin-beam bending locking error from 85.9 % (plain hex8) to $< 10^{-12}$.

5 Constrained Quality-Diversity Search (FI-2Pop + Multi-Emitter Bandit)

crates/g4/src/search.rs implements a deterministic, batch-synchronous constrained QD engine:

- **Lloyd-relaxed 5D CVT archive (FI-2Pop):** Niche centroids in the 5D phenotypic descriptor space (finished mass m, blade reach R, claw turn angle $\Delta\theta$, grip H/T, and hollowness fraction η_{void}) are computed via deterministic Lloyd relaxation [2]. Each cell maintains both a *feasible elite* ($V(x) = 0$, maximizing $W_{\text{ERGOLIGHT_v2}}$) and an *infeasible stepping stone* ($V(x) > 0$, minimizing $V(x)$) [5], plus a local Pareto front over ($m_{\text{finished}}, -E, -\min$ margin).
- **UCB1 four-emitter bandit:** Offspring batches are generated by four specialized emitters scheduled by UCB1: (1) Sep-CMA-MAE [3] with diagonal covariance adaptation in $[0, 1]^{130}$, (2) a **Lamarckian constraint-repair emitter** that projects violating genes onto the active structural/geometric boundary, (3) a **feasible mass-descent emitter** that applies coordinated mass-saving steps along Levers L-1–L-8 whenever safety margins exceed 1.15, and (4) a module-aware Iso+LineDD crossover emitter.
- **Periodic $h = 0.20$ mm re-certification and cross-arch determinism:** Top archive elites are periodically re-certified at Tier 2 ($h = 0.20$ mm, 3-level plate GCI, and 3D TET10 tip FE). All transcendental math uses pure-Rust libm and parallel batches are reduced in deterministic index order, guaranteeing bit-identical results across thread counts (T-DET-1).

Table 2 | Eight-lever mass reduction waterfall (evidence/g4/studies/e05_mass_levers.json)

Step	Architectural Lever Applied	Mass (g)	Δm (g)	Sc
L-0	Uniform-skin unperforated baseline	69.56	—	72
L-1	Flange-graded grip skin (0.80 mm flanks, +0.42 mm chords)	64.96	−4.60	75
L-2	Perforated Warren web (62 % circular cutouts)	62.66	−2.30	77
L-3	Lamarckian stress-graded Sheet-TPMS core	59.30	−3.36	79
L-4	Drained hollow ring gallery + 32.0 mm ring OD	54.44	−4.86	83
L-5	Bi-concave blade fuller + lean 73.5 mm chord reach	49.41	−5.03	86
L-6	Decoupled hollow blade core ($\rho_{\text{blade}} = 0.13$)	46.94	−2.47	88
L-7	Lean anthropometric grip + zero-mean texture + Mod. T	42.66	−4.28	91
L-8	Distal tip-bisector diamond crown (+0.38 mm)	**42.85**	+0.19	**92**

6 Experimental Results, Mass Waterfall, and Certified Winner

6.1 Eight-Lever Mass Waterfall (Study E-05)

Table 2 traces the sequential application of the eight structural mass-reduction levers (L-1–L-8) starting from a uniform-skin, unperforated baseline (L-0, 69.56 g finished). Together, flange-graded grip skins (L-1), perforated Warren webs (L-2), stress-graded TPMS (L-3), the drained hollow ring gallery (L-4), the bi-concave blade fuller (L-5), decoupled hollow blade core (L-6), lean anthropometric grip envelope with zero-mean texture and trabecular flank windows (L-7), and the distal tip-bisector crown (L-8) reduce finished mass by **26.71 g** (−38.4 %, down to **42.85 g**) while maintaining positive safety margins across all 29 gates.

6.2 Multi-Seed Cohort and Five-Arm Ablation Study (E0–E5)

Table 3 summarizes the five-arm experimental cohort executed by g4-evolve (evidence/g4/v4.0.0/evolution_summary.json). Full G4 (E3, 5 seeds) achieves a median score of **91.84** (IQR 0.13) and median finished mass of **37.01 g**, outperforming the no-Module-T ablation (E5, median 92.27, mass 35.91 g), single-population penalty CMA (E2, median 88.73), legacy W_ERGOLIGHT_v1 search (E1, median 91.90), and pure scrambled Sobol search (E0, median 88.08) with Vargha–Delaney effect size $\hat{A}_{12} = 1.00$ against E0/E1/E2.

Table 3 | Multi-seed cohort and ablation comparison (evidence/g4/v4.0.0/evolution_summary.json)

Experiment Arm	Seeds	Med. Score	IQR	Med. Mass (g)	$\hat{A}_{12}$ vs. E3
E3: G4 Full (FI-2Pop + Mod. T)	5	**91.84**	0.13	**37.01**	0.50
E5: G4 Core (No Module T)	3	92.27	0.34	35.91	0.93
E2: Single-Pop Penalty CMA	3	88.73	0.62	44.12	1.00
E1: Legacy W_ERGOLIGHT_v1	3	91.90	1.15	45.80	0.00
E0: Pure Scrambled Sobol	3	88.08	2.40	48.95	1.00

Table 4 | Head-to-head comparison: G3 v3.0.0 winner vs. certified G4 v4.0.0 winner `g4v400_s042_62763aa2`

Metric / Certification Gate	Req. (v4)	G3 v3.0.0	G4 v4.0.0
Finished CAD mass (g)	< 50.0	67.97	**35.42**
As-printed AM build mass (g)	—	71.09	**35.68**
Solid envelope mass (g) / Hollowness	—	101.0 / 29.6 %	**108.30 / 67.3 %**
Ring / Grip / Guard / Blade mass (g)	—	9.8 / 37.9 / 4.2 / 19.2	**3.67 / 19.94 / 1.74 / 10.07**
Score (W_ERGOLIGHT_v2 / legacy v1)	≥ 50.0	< 50 (infeas.) / 80.78	**92.45 / 94.90**
Planform point angle at 3 mm (°)	≤ 55.0°	92.22°	**41.38°**
Roache 3-level plate GCI_{21} (%)	≤ 5.0 %	uncomputed	**0.37 %**
Hiker lateral 180 N SF_{LCB}	≥ 1.35	4.54	**4.62**
Hunter lateral 260 N SF_{LCB}	≥ 1.15	3.14	**3.20**
Body torsion 1.82 N·m SF_{LCB}	≥ 1.25	15.65	**14.74**
Human-max torsion 16.2 N·m SF_{LCB}	≥ 1.10	1.76	**1.66**
Solid-tip torsion 0.80 N·m SF_{LCB}	≥ 1.25	0.431 (1.82 N·m)	**3.29**
Solid-tip lateral pry 60 N SF_{LCB}	≥ 1.20	—	**32.21**
Spine push 500 N SF_{LCB}	≥ 1.50	6.22	**27.10**
Spine baton 3,427 N ultimate SF_{LCB}	≥ 1.10	0.515 (ring clamp)	**4.27**
Ring bearing 300 N / back 30 N SF_{LCB}	≥ 2.00	5.66 / 3.92	**20.10 / 67.01**
Grip bilateral squeeze 250 N SF_{LCB}	≥ 1.50	—	**8.19**
Serration tooth snag+pry SF_{LCB} (checked)	≥ 1.25	2.13 (4/5, 5/6)	**12.07** (4/4, 5/5)
Goodman high-cycle fatigue 180 N SF_{LCB}	≥ 1.10	1.84 (true)	**2.31**
Lateral-torsional buckling λ_{LCB}	≥ 2.00	3.85	**31.19**
Combined 70 % lat. + 70 % twist SF_{LCB}	≥ 1.15	—	**4.47**
Per-case failure sequence $\min \text{ratio}_{LCB}$	≥ 1.10	0.253 (spine)	**2.17**
TPMS face-sheet wrinkling SF_{LCB}	≥ 1.35	—	**63.68**
CoM distance to index choil (mm)	≤ 15.0	11.40	**3.63**

6.3 The Certified Sub-50 g Winner (`g4v400_s042_62763aa2`)

Table 4 compares the certified G4 v4.0.0 winner `g4v400_s042_62763aa2` (genome hash `62763aa2`) against the G3 v3.0.0 winner hz2i012_g00609_d55af923. The G4 winner weighs **35.42 g** as finished (**35.68 g** as printed with the +0.30 mm bevel grind allowance, a **38.4 % mass reduction** vs. G3's 67.97 g finished / 71.09 g printed meshes), partitioned across ring 3.67 g, grip 19.94 g, guard 1.74 g, and blade 10.07 g (root 2.98 g, mid 4.32 g, tip 2.78 g). Every one of the 17 structural LCB safety factors exceeds its required threshold ($V(x) = 0$).

7 Scope, Limitations, and Reproducibility

Like G3, G4 is a computational engineering release: physical L-PBF fabrication, hot isostatic pressing (HIP), CNC bevel grinding, and destructive mechanical testing remain required before operational qualification. All code, study receipts (evid ence/g4/studies/), cohort telemetry (evidence/g4/v4.0.0/), and the six-file manufacturing upload kit (cad/) are deterministically reproducible via cargo test --manifest-path crates/g4/Cargo.t oml --release and g4-studies, g4-evolve, g4-release-tex, and g4-export.

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